



White Band Associates

WHITE BAND ASSOCIATES

Cyber Security

Summer Internship 2026

Networking Task 01

Name : Bhakti Mahadev Parhad

Domain : Cyber Security

Week 1 - Networking

INDEX

Sr. No.	Contents	Page No.
1.	Introduction	3
2.	Objective	4
3.	System Information	5
4.	Network Configuration Details	6
5.	Networking Concepts	8
6.	Network Diagram	10
7.	Ping Test	11
8.	Traceroute Test	12
9.	Linux (Kali) Verification	13
10.	Learning Outcomes	16
11.	Conclusion	17

INTRODUCTION

Computer networking is the process of connecting devices to enable communication and data exchange. It forms the foundation of modern technology, allowing systems to access the internet, share resources, and communicate efficiently. Understanding networking concepts such as IP addresses, MAC addresses, DNS, and network connectivity is essential for cybersecurity professionals. This task focuses on exploring basic network configuration and performing connectivity tests using networking tools and commands.

OBJECTIVE

The objective of this task is to understand basic networking concepts and gain practical experience in identifying network configuration details such as IP Address, MAC Address, Default Gateway, and DNS Server. This task also aims to verify network connectivity using Ping and Traceroute commands and develop a fundamental understanding of network communication.

SYSTEM INFORMATION

Parameter	Details
Name	Bhakti Parhad
Operating System	Windows 11
Virtualization Software	VMware Workstation
Linux Distribution	Kali Linux
Network Adapter	Intel(R) Wi-Fi 6 AX203
Task Domain	Networking
Internship Program	White Band Associates Summer Internship 2026

NETWORK CONFIGURATION DETAILS

The following network configuration details were obtained using the ipconfig /all command in Windows Command Prompt.

Parameter	Value
Hostname	LAPTOP-UPABTK5H
IPv4 Address	10.180.51.70
MAC Address (Physical Address)	XX-XX-XX-XX-XX-XX
Default Gateway	10.180.51.197
DNS Server	10.180.51.197

Description :

1. **Hostname** : Identifies the computer on the network.
2. **IPv4 Address** : The unique address assigned to the device for communication within the network.
3. **MAC Address** : A unique hardware address assigned to the network interface card.
4. **Default Gateway** : The router that forwards network traffic to external networks and the internet.
5. **DNS Server** : Translates domain names such as google.com into IP addresses.

Figure 1 : Hostname Information

```
C:\Users\bhakt>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : LAPTOP-UPABTK5H
    Primary Dns Suffix . . . . . :
    Node Type . . . . . :
    IP Routing Enabled. . . . . :
    WINS Proxy Enabled. . . . . :
```

Figure 2: IPv4 Address and MAC Address

```
Connection-specific DNS Suffix . :
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1
Physical Address. . . . . :
DHCP Enabled. . . . . :
Autoconfiguration Enabled . . . . :
Link-local IPv6 Address . . . . . :
IPv4 Address. . . . . : 192.168.27.1(Preferred)
Subnet Mask . . . . . :
Lease Obtained. . . . . :
Lease Expires . . . . . :
Default Gateway . . . . . :
DHCP Server . . . . . :
DHCPv6 IAID . . . . . :
DHCPv6 Client DUID. . . . . :
```

Figure 3: Default Gateway and DNS Servers

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    Description . . . . . :
    Physical Address. . . . . :
    DHCP Enabled. . . . . :
    Autoconfiguration Enabled . . . . :
    IPv6 Address. . . . . :
    Temporary IPv6 Address. . . . . :
    Link-local IPv6 Address . . . . . :
    IPv4 Address. . . . . :
    Subnet Mask . . . . . :
    Lease Obtained. . . . . :
    Lease Expires . . . . . :
    Default Gateway . . . . . : fe80::147d:a7ff:fe22:b497%10
                                10.180.51.197
    DHCP Server . . . . . :
    DHCPv6 IAID . . . . . :
    DHCPv6 Client DUID. . . . . :
    DNS Servers . . . . . : 10.180.51.197
                                2409:40c2:6420:17c2::de
    NetBIOS over Tcpip. . . . . : Enabled
```

NETWORKING CONCEPTS

What is an IP Address?

An *IP (Internet Protocol)* Address is a unique numerical identifier assigned to a device on a network. It enables communication between devices and helps in sending and receiving data.

What is a MAC Address?

A *MAC (Media Access Control)* Address is a unique hardware address assigned to a *network interface card (NIC)*. It is used to identify devices within a local network.

What is a Default Gateway?

A Default Gateway is a network device, usually a router, that forwards traffic from a local network to external networks such as the Internet.

What is a DNS Server?

A *DNS (Domain Name System)* Server translates human-readable domain names such as google.com into IP addresses that computers can understand.

What is a Public IP Address?

A Public IP Address is assigned by an Internet Service Provider (ISP) and is used for communication over the Internet.

What is a Private IP Address?

A Private IP Address is used within a local network and cannot be accessed directly from the Internet. *Example : 10.180.51.70.*

What is Ping?

Ping is a network diagnostic tool used to test connectivity between two devices and measure response time.

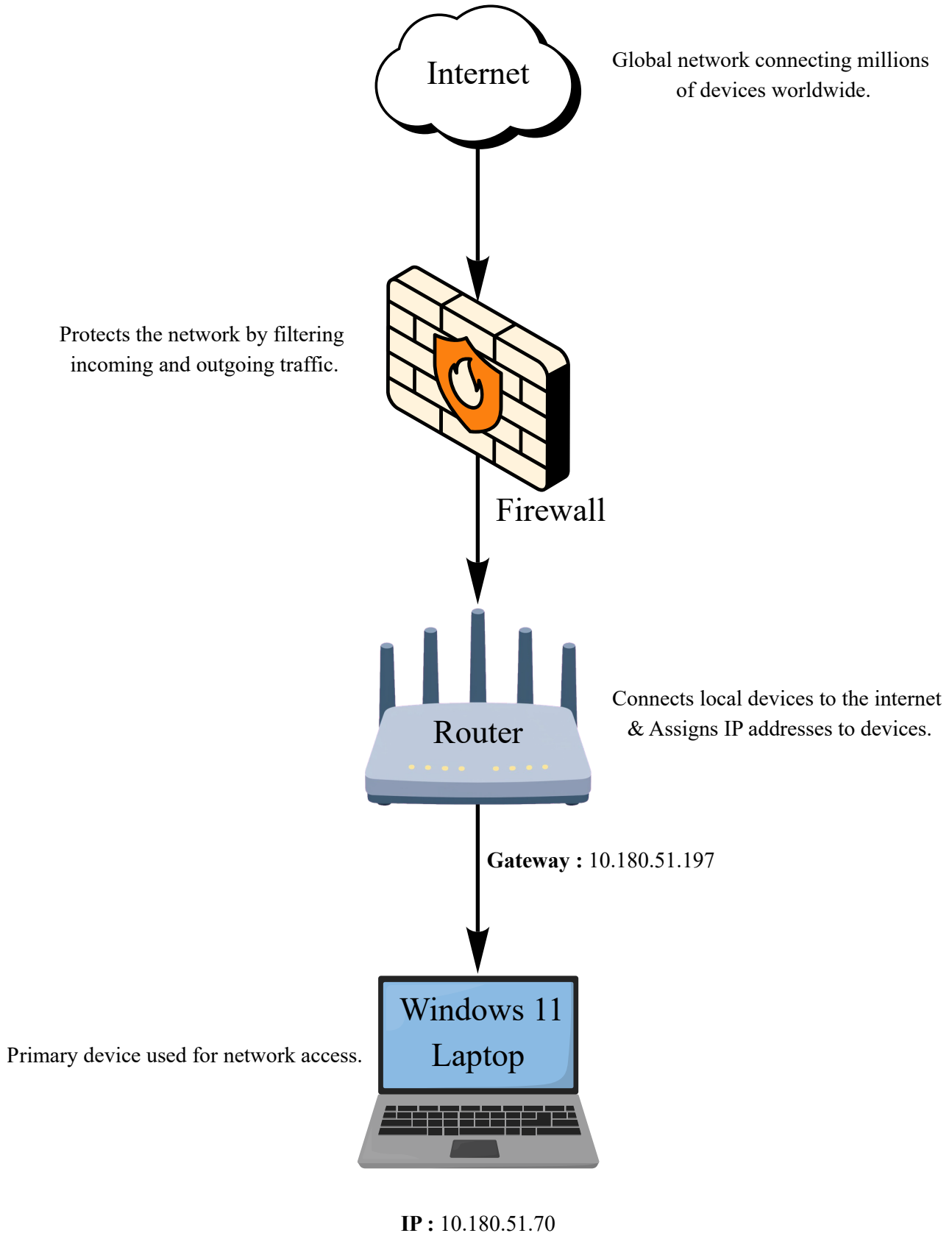
What is Traceroute?

Traceroute is a network diagnostic tool used to identify the path that data packets take from the source device to the destination.

Difference between Public IP and Private IP.

Public IP Address	Private IP Address
Used for communication over the Internet.	Used within a local network (LAN).
Accessible from anywhere on the Internet.	Not directly accessible from the Internet.
Assigned by an Internet Service Provider (ISP).	Assigned by a router or local network administrator.
Must be unique globally.	Can be reused in different private networks.
Example: 8.8.8.8, 1.1.1.1	Example: 192.168.1.10, 10.0.0.5

NETWORK DIAGRAM



PING TEST

Command Used

ping google.com

Purpose

The Ping command is used to test connectivity between the local device and a remote host. It helps determine whether the destination is reachable and measures the response time.

Observation

The command successfully received replies from Google's server, indicating that the internet connection was active and functioning correctly. The response time was displayed in milliseconds (ms), confirming successful communication with the destination host.

Figure 4 : Ping Command Output

```
C:\Users\bhakt>ping google.com

Pinging google.com [2404:6800:4002:80b::200e] with 32 bytes of data:
Reply from 2404:6800:4002:80b::200e: time=70ms
Reply from 2404:6800:4002:80b::200e: time=85ms
Reply from 2404:6800:4002:80b::200e: time=61ms
Reply from 2404:6800:4002:80b::200e: time=69ms

Ping statistics for 2404:6800:4002:80b::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 61ms, Maximum = 85ms, Average = 71ms
```

TRACEROUTE TEST

Command Used

tracert google.com

Purpose

The Traceroute command is used to identify the path taken by network packets from the source device to the destination server. It displays the intermediate routers (hops) through which the packets travel.

Observation

The command successfully displayed multiple hops between the local system and Google's server. Each hop represented a router or network device through which the packets passed before reaching the destination.

Figure 5 : Traceroute Command Output

```
C:\Users\bhakt>tracert google.com

Tracing route to google.com [2404:6800:4002:80b::200e]
over a maximum of 30 hops:

  1    3 ms    3 ms    40 ms  2409:40c2:6420:17c2::de
  2    *      *      *      Request timed out.
  3   70 ms   35 ms   38 ms  2405:200:5205:28:3925::ff21
  4    *      *      *      Request timed out.
  5    *      *      *      Request timed out.
  6   70 ms   30 ms   35 ms  2405:200:801:6000::16
  7    *      *      *      Request timed out.
  8    *      *      *      Request timed out.
  9   57 ms   40 ms   71 ms  2404:6800:8281:2c0::1
 10   78 ms   93 ms   43 ms  2404:6800:8281:2c0::1
 11   48 ms   54 ms   64 ms  2001:4860:0:1::269e
 12   61 ms   37 ms   43 ms  2001:4860:0:1::7140
 13   77 ms   34 ms   38 ms  2001:4860::c:4004:53ca
 14   75 ms   70 ms   55 ms  2001:4860::c:4004:153f
 15   76 ms   88 ms   91 ms  2001:4860:0:1::75b6
 16   98 ms   72 ms   76 ms  tzdela-bb-in-x0e.1e100.net [2404:6800:4002:80b::200e]

Trace complete.
```

LINUX (KALI) VERIFICATION

Objective

To verify network configuration and network connectivity using networking commands in Kali Linux.

Commands Used

- `ip addr`
- `ping google.com`
- `traceroute google.com`

Purpose

- *ip addr* was used to view network interface details, IP addresses, and MAC addresses.
- *ping* was used to verify connectivity between the Kali Linux system and Google's server.
- *traceroute* was used to identify the path taken by packets from the Kali Linux system to the destination server.

Observation

The Kali Linux virtual machine successfully displayed its network configuration, including the assigned IP address and network interface information. The ping test received successful replies from Google's server, confirming active internet connectivity. The traceroute command displayed the network path and intermediate hops between the source and destination.

Figure 6 : Kali Linux Network Configuration and Ping Test

```
kali@kali ~  
$ ip addr  
1:   
  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
  
(kali@kali)~  
$ ping google.com  
PING google.com (2404:6800:4002:82a::200e) 56 data bytes  
64 bytes from tzdelb-ay-in-x0e.1e100.net (2404:6800:4002:82a::200e): icmp_seq=1 ttl=116 time=187 ms  
64 bytes from tzdelb-ay-in-x0e.1e100.net (2404:6800:4002:82a::200e): icmp_seq=2 ttl=116 time=95.7 ms  
^C  
--- google.com ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1532ms  
rtt min/avg/max/mdev = 95.666/141.486/187.306/45.820 ms
```

Figure 7 : Kali Linux Traceroute Verification

```
kali@kali ~  
$ traceroute google.com  
traceroute to google.com (2404:6800:4002:82a::200e), 30 hops max, 80 byte packets  
1 2409:40c2:6425:ab07::73 (2409:40c2:6425:ab07::73) 19.718 ms 18.381 ms 18.092 ms  
2 * * *  
3 2405:200:5205:28:3925::ff21 (2405:200:5205:28:3925::ff21) 267.351 ms 237.774 ms 237.673 ms  
4 * * *  
5 * * *  
6 2405:200:801:6000::16 (2405:200:801:6000::16) 269.187 ms 51.734 ms 64.720 ms  
7 * * *  
8 * * *  
9 2001:4860:1:1::a7c (2001:4860:1:1::a7c) 40.944 ms 2001:4860:1:1::1022 (2001:4860:1:1::1022) 30.964 ms 39.722 ms  
10 2404:6800:8281:1c0::1 (2404:6800:8281:1c0::1) 60.079 ms 2404:6800:8281:300::1 (2404:6800:8281:300::1) 465.060 ms 2404:6800:8281:300::1 (2404:6800:8281:300::1) 57.786 ms  
11 2001:4860:0:1::5ed4 (2001:4860:0:1::5ed4) 100.666 ms 2001:4860:0:1::443a (2001:4860:0:1::443a) 436.171 ms 2001:4860:0:1::7976 (2001:4860:0:1::7976) 70.276 ms  
12 2001:4860::c:4004:53be (2001:4860::c:4004:53be) 73.784 ms 2001:4860:0:1::3fe4 (2001:4860:0:1::3fe4) 49.589 ms 2001:4860:0:1::876a (2001:4860:0:1::876a) 84.938 ms  
13 2001:4860::c:4004:2137 (2001:4860::c:4004:2137) 72.837 ms 2001:4860::c:4004:53bf (2001:4860::c:4004:53bf) 94.917 ms 96.194 ms  
14 2001:4860:0:1::77d0 (2001:4860:0:1::77d0) 93.806 ms 2001:4860::c:4004:153f (2001:4860::c:4004:153f) 87.194 ms 182.843 ms  
15 2001:4860::9:4001:d9e7 (2001:4860::9:4001:d9e7) 131.152 ms 2001:4860:0:1::77d0 (2001:4860:0:1::77d0) 119.616 ms 2001:4860:0:1::9:4001:d9e7 (2001:4860:0:1::9:4001:d9e7) 72.608 ms  
16 2001:4860:0:1::53a1 (2001:4860:0:1::53a1) 82.193 ms 2001:4860:0:1::539f (2001:4860:0:1::539f) 81.961 ms 2001:4860:0:1::7:4860:0:1::77dd (2001:4860:0:1::77dd) 71.850 ms  
17 tzdelb-ay-in-x0e.1e100.net (2404:6800:4002:82a::200e) 79.849 ms 2001:4860:0:1::53a1 (2001:4860:0:1::53a1) 85.568 ms 115
```

Question & Answers :

1.Was the ping successful?

Yes, the ping was successful. Replies were received from the destination server, indicating that network connectivity was working properly.

2.How many hops were shown?

Windows :

The traceroute command showed 16 hops between the source device and the destination server.

Kali Linux :

The traceroute command showed 17 hops between the source device and the destination server.

3.What is the purpose of traceroute?

Traceroute is used to identify the path taken by data packets from a source device to a destination. It helps analyze network routes and troubleshoot connectivity or network delay issues.

LEARNING OUTCOMES

Through this task, I gained practical knowledge of basic networking concepts and commands. The key learning outcomes are :

- Learned how to identify network configuration details using the `ipconfig /all` command.
- Understood the purpose and importance of IP Addresses, MAC Addresses, DNS Servers, and Default Gateways.
- Learned the difference between Public and Private IP Addresses.
- Performed network connectivity testing using the `ping` command.
- Analyzed the path taken by network packets using the `tracert` command.
- Created a basic network diagram to understand network communication flow.
- Gained hands-on experience with networking tools in both Windows and Kali Linux environments.
- Improved documentation and reporting skills by organizing findings and screenshots in a structured format.

CONCLUSION

This task provided practical exposure to fundamental networking concepts and tools. By examining network configuration details, performing connectivity tests using Ping and Traceroute, and creating a network diagram, I gained a better understanding of how devices communicate within a network and access the Internet. The task also helped strengthen my knowledge of IP addressing, DNS, MAC addresses, and network troubleshooting, which are essential foundations for Cyber Security and IT.